① Să se calculeze derivatele parțiale de ordinul I și II în punctul $(0,0)$ pentru $f(x,y)=\sqrt{1-xy}\,e^{2y}$

$$\frac{\partial f}{\partial x}=\frac{1}{2\sqrt{1-xy}}\cdot(-y)\cdot e^{2y}=\frac{-y}{2}\cdot(1-xy)^{-1/2}e^{2y}$$

$$\frac{\partial f}{\partial y}=\frac{1}{2\sqrt{1-xy}}\cdot(-x)e^{2y}+\sqrt{1-xy}\cdot e^{2y}\cdot 2=\frac{-x}{2\sqrt{1-xy}}e^{2y}+2\sqrt{1-xy}\,e^{2y}$$

$$\frac{\partial^2 f}{\partial x^2}=\frac{-y}{2}\cdot\left(-\frac{1}{2}\right)\cdot(1-xy)^{-3/2}\cdot(-y)\cdot e^{2y}=\frac{y^2}{4}(1-xy)^{-3/2}e^{2y}$$

$$\frac{\partial^2 f}{\partial y^2}=\frac{-x}{2}\cdot\left(-\frac{1}{2}\right)\cdot(1-xy)^{-3/2}\cdot(-x)\cdot e^{2y}+\frac{-x}{2\sqrt{1-xy}}\cdot e^{2y}\cdot 2+2\cdot\frac{-x}{2\sqrt{1-xy}}\cdot e^{2y}$$
$$+2\sqrt{1-xy}\cdot e^{2y}\cdot 2$$

$$\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial^2 f}{\partial y\partial x}=\frac{-1}{2}(1-xy)^{-1/2}e^{2y}+\frac{-y}{2}\cdot\left(-\frac{1}{2}\right)(1-xy)^{-3/2}\cdot(-x)\cdot e^{2y}+$$
$$+\frac{-y}{\cancel{2}}(1-xy)^{-1/2}e^{2y}\cdot\cancel{2}$$
$$=\frac{-e^{2y}}{2}(1-xy)^{-1/2}+\frac{-xy\,e^{2y}}{4}(1-xy)^{-3/2}-y\,e^{2y}(1-xy)^{-1/2}$$

$\frac{\partial f}{\partial x}(0,0)=0$

$\frac{\partial f}{\partial y}(0,0)=0+2\cdot\sqrt{1}\cdot e^0=2$

$\frac{\partial^2 f}{\partial x^2}(0,0)=0$

$\frac{\partial^2 f}{\partial y^2}(0,0)=0+0+0+4\sqrt{1}\cdot e^0=4$

$\frac{\partial^2 f}{\partial x\partial y}(0,0)=\frac{-e^0}{2}\cdot 1+0-0=\frac{-1}{2}$

② Calculați Δf, $\Delta^2 f$ pentru $f(x,y)=\operatorname{arctg}\frac{x+y}{1-xy}$

$$\Delta f=\frac{\partial f}{\partial x}dx+\frac{\partial f}{\partial y}dy$$

$$\frac{\partial f}{\partial x}=\frac{1}{1+\left(\frac{x+y}{1-xy}\right)^2}\cdot\frac{1(1-xy)-(x+y)\cdot(-y)}{(1-xy)^2}=$$
$$=\frac{\cancel{(1-xy)^2}}{1-2xy+x^2y^2+x^2+2xy+y^2}\cdot\frac{1-xy+xy+y^2}{\cancel{(1-xy)^2}}=\frac{1+y^2}{1+x^2+y^2+x^2y^2}$$
$$=\frac{1+y^2}{1+y^2+x^2(1+y^2)}=\frac{\cancel{1+y^2}}{(1+x^2)\cancel{(1+y^2)}}=\frac{1}{1+x^2}$$

$$\frac{\partial f}{\partial y}=\frac{1}{1+\left(\frac{x+y}{1-xy}\right)^2}\cdot\frac{1(1-xy)-(x+y)\cdot(-x)}{(1-xy)^2}=\frac{\cancel{(1-xy)^2}}{1-2xy+x^2y^2+x^2+2xy+y^2}$$
$$\cdot\frac{1-xy+x^2+xy}{\cancel{(1-xy)^2}}=\frac{1+x^2}{1+x^2+y^2(x^2+1)}=\frac{\cancel{1+x^2}}{\cancel{(1+x^2)}(1+y^2)}=\frac{1}{1+y^2}$$

$$\Delta f=\frac{1}{1+x^2}dx+\frac{1}{1+y^2}dy$$

$$\Delta^2 f=\frac{\partial^2 f}{\partial x^2}dx^2+2\cdot\frac{\partial^2 f}{\partial x\partial y}dx\,dy+\frac{\partial^2 f}{\partial y^2}dy^2$$

$$\frac{\partial^2 f}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{1}{1+x^2}\right)=-1(1+x^2)^{-2}\cdot 2x=\frac{-2x}{(1+x^2)^2}$$

$$\frac{\partial^2 f}{\partial y^2}=\frac{-2y}{(1+y^2)^2}$$

$$\frac{\partial^2 f}{\partial x\partial y}=0$$

$$\Delta^2 f=\frac{-2x}{(1+x^2)^2}dx^2+\frac{-2y}{(1+y^2)^2}dy^2$$

③ Calculați $\Delta f(1,1,0)$ pentru $f(x,y,z)=\left(\sqrt{4y^2+z^2+1},\ \ln(x^2+2+3z^4)\right)$

$\frac{\partial f_1}{\partial x}=0$ $\qquad$ $\frac{\partial f_2}{\partial x}=\frac{2x}{x^2+2+3z^4}$

$\frac{\partial f_1}{\partial y}=\frac{4\cdot\cancel{2}y}{\cancel{2}\sqrt{4y^2+z^2+1}}=\frac{4y}{\sqrt{4y^2+z^2+1}}$ $\qquad$ $\frac{\partial f_2}{\partial y}=0$

$\frac{\partial f_1}{\partial z}=\frac{2z}{2\sqrt{4y^2+z^2+1}}=\frac{z}{\sqrt{4y^2+z^2+1}}$ $\qquad$ $\frac{\partial f_2}{\partial z}=\frac{12z^3}{x^2+2+3z^4}$

$$J_f(x,y,z)=\begin{pmatrix}0 & \frac{4y}{\sqrt{4y^2+z^2+1}} & \frac{z}{\sqrt{4y^2+z^2+1}}\\ \frac{2x}{x^2+2+3z^4} & 0 & \frac{12z^3}{x^2+2+3z^4}\end{pmatrix}$$

$$D_f(x,y,z)=J_f(x,y,z)\cdot\begin{pmatrix}dx\\dy\\dz\end{pmatrix}=\begin{pmatrix}0 & \frac{4y}{\sqrt{4y^2+z^2+1}} & \frac{z}{\sqrt{4y^2+z^2+1}}\\ \frac{2x}{x^2+2+3z^4} & 0 & \frac{12z^3}{x^2+2+3z^4}\end{pmatrix}\begin{pmatrix}dx\\dy\\dz\end{pmatrix}$$
$$=\begin{pmatrix}\frac{4y}{\sqrt{4y^2+z^2+1}}dy+\frac{z}{\sqrt{4y^2+z^2+1}}dz\\ \frac{2x}{x^2+2+3z^4}dx+\frac{12z^3}{x^2+2+3z^4}dz\end{pmatrix}$$

④ Polinomul Taylor de gradul 2 în $(1,0)$ pentru $f(x,y)=\sqrt{x^2+y^2}$

$$T_2(x,y)=f(x_0,y_0)+\frac{1}{1!}\left[(x-x_0)\frac{\partial f}{\partial x}(x_0,y_0)+(y-y_0)\frac{\partial f}{\partial y}(x_0,y_0)\right]+$$
$$+\frac{1}{2!}\left[(x-x_0)^2\frac{\partial^2 f}{\partial x^2}(x_0,y_0)+2(x-x_0)(y-y_0)\frac{\partial^2 f}{\partial x\partial y}(x_0,y_0)+(y-y_0)^2\frac{\partial^2 f}{\partial y^2}(x_0,y_0)\right]$$

$$\frac{\partial f}{\partial x}=\frac{2x}{2\sqrt{x^2+y^2}}=\frac{x}{\sqrt{x^2+y^2}}=x(x^2+y^2)^{-1/2}$$

$$\frac{\partial^2 f}{\partial x^2}=\frac{\sqrt{x^2+y^2}-x\cdot\frac{1}{\cancel{2}\sqrt{x^2+y^2}}\cdot\cancel{2}x}{x^2+y^2}=\frac{x^2+y^2-x^2}{(x^2+y^2)^{3/2}}=\frac{y^2}{(x^2+y^2)^{3/2}}$$

$$\frac{\partial f}{\partial y}=\frac{y}{\sqrt{x^2+y^2}}\qquad f(1,0)=\sqrt{1^2+0}=1$$

$$\frac{\partial^2 f}{\partial y^2}=\frac{x^2}{(x^2+y^2)^{3/2}}$$

$$\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial^2 f}{\partial y\partial x}=y\cdot\frac{-2x}{2(x^2+y^2)^{-3/2}}=\frac{-xy}{(x^2+y^2)^{-3/2}}$$

$\frac{\partial f}{\partial x}(1,0)=1\cdot 1=1$ $\qquad$ $\frac{\partial^2 f}{\partial x^2}(1,0)=0$ $\qquad$ $\frac{\partial^2 f}{\partial x\partial y}=0$

$\frac{\partial f}{\partial y}(1,0)=0$ $\qquad$ $\frac{\partial^2 f}{\partial y^2}(1,0)=\frac{1}{1}=1$

$$T_2(x,y)=1+\frac{1}{1!}\left[(x-1)\cdot 1+(y-0)\cdot 0\right]+\frac{1}{2!}\left[(x-1)^2\cdot 0+2(x-1)(y-0)\cdot 0\right.$$
$$\left.+(y-0)^2\,1\right]$$

$$T_2(x,y)=1+x-1+\frac{1}{2}y^2=x+\frac{y^2}{2}$$

⑤ Să se calculeze aproximativ cu ajutorul pol. Taylor de grad 2 expresia $E=1{,}03^2\cdot e^{0{,}5}\sqrt[3]{1{,}1}$ $\quad f(x,y,z)=x^2e^y\sqrt[3]{z}$

$$f(x_0+h,y_0+k,z_0+l)\simeq f(x_0,y_0,z_0)+\frac{1}{1!}\left[h\frac{\partial f}{\partial x}(x_0,y_0,z_0)+k\frac{\partial f}{\partial y}(x_0,y_0,z_0)+\right.$$
$$\left.l\frac{\partial f}{\partial z}(x_0,y_0,z_0)\right]+\frac{1}{2!}\left[h^2\frac{\partial^2 f}{\partial x^2}(x_0,y_0,z_0)+k^2\frac{\partial^2 f}{\partial y^2}(x_0,y_0,z_0)+l^2\frac{\partial^2 f}{\partial z^2}(x_0,y_0,z_0)\right.$$
$$\left.+2hk\frac{\partial^2 f}{\partial x\partial y}(x_0,y_0,z_0)+2hl\frac{\partial^2 f}{\partial x\partial z}(x_0,y_0,z_0)+2kl\frac{\partial^2 f}{\partial y\partial z}(x_0,y_0,z_0)\right]$$

$x_0=1$ $\qquad$ $h=0{,}03$

$y_0=0$ $\qquad$ $k=0{,}5$

$z_0=1$ $\qquad$ $l=0{,}1$

$\frac{\partial f}{\partial x}=2xe^y\sqrt[3]{z}$ $\qquad$ $\frac{\partial^2 f}{\partial x^2}=2e^y\sqrt[3]{z}$ $\qquad$ $\frac{\partial^2 f}{\partial x\partial y}=2xe^y\sqrt[3]{z}$

$\frac{\partial f}{\partial y}=x^2\cdot e^y\sqrt[3]{z}$ $\qquad$ $\frac{\partial^2 f}{\partial y^2}=x^2e^y\sqrt[3]{z}$ $\qquad$ $\frac{\partial^2 f}{\partial x\partial z}=2xe^y\cdot\frac{1}{3\sqrt[3]{z^2}}$

$\frac{\partial f}{\partial z}=x^2e^y\cdot\frac{1}{3\sqrt[3]{z^2}}$ $\qquad$ $\frac{\partial^2 f}{\partial z^2}=x^2e^y\cdot\frac{1}{3}\left(-\frac{2}{3}\right)\cdot z^{-5/3}$ $\qquad$ $\frac{\partial^2 f}{\partial y\partial z}=x^2e^y\frac{1}{3\sqrt[3]{z^2}}$

$\frac{\partial f}{\partial x}(1,0,1)=2\cdot 1\cdot e^0\cdot 1=2$ $\qquad$ $\frac{\partial^2 f}{\partial x^2}(1,0,1)=2$ $\qquad$ $\frac{\partial^2 f}{\partial x\partial y}(1,0,1)=2$

$\frac{\partial f}{\partial y}(1,0,1)=1$ $\qquad$ $\frac{\partial^2 f}{\partial y^2}(1,0,1)=1$ $\qquad$ $\frac{\partial^2 f}{\partial x\partial z}(1,0,1)=\frac{2}{3}$

$\frac{\partial f}{\partial z}(1,0,1)=\frac{1}{3}$ $\qquad$ $\frac{\partial^2 f}{\partial z^2}(1,0,1)=-\frac{2}{9}$ $\qquad$ $\frac{\partial^2 f}{\partial y\partial z}(1,0,1)=\frac{1}{3}$

$$f(1{,}03;0{,}5;1{,}1)\simeq f(1,0,1)+\left[0{,}03\cdot 2+0{,}5\cdot 1+0{,}1\cdot\frac{1}{3}\right]+\frac{1}{2}\left[0{,}03^2\cdot 2+0{,}5^2\cdot 1+\right.$$
$$\left.0{,}1^2\cdot\left(-\frac{2}{9}\right)+2\cdot 0{,}03\cdot 0{,}5\cdot 2+2\cdot 0{,}03\cdot 0{,}1\cdot\frac{2}{3}+2\cdot 0{,}5\cdot 0{,}1\cdot\frac{1}{3}\right]$$

Algoritm de determinare a extremelor locale pentru funcții de 3 variabile $f(x,y,z)$

(1) Se determină punctele critice din sistemul $\begin{cases}\frac{\partial f}{\partial x}=0\\ \frac{\partial f}{\partial y}=0\\ \frac{\partial f}{\partial z}=0\end{cases}$

Fie $M_0(x_0,y_0,z_0)$ un pct. critic

(2) Se calculează deriv. parțiale de ordinul 2 ale în punctul M_0

(3) Se scrie matricea Hessiană

$$H(M_0)=\begin{pmatrix}\frac{\partial^2 f}{\partial x^2}(M_0) & \frac{\partial^2 f}{\partial x\partial y}(M_0) & \frac{\partial^2 f}{\partial x\partial z}(M_0)\\ \frac{\partial^2 f}{\partial y\partial x}(M_0) & \frac{\partial^2 f}{\partial y^2}(M_0) & \frac{\partial^2 f}{\partial y\partial z}(M_0)\\ \frac{\partial^2 f}{\partial z\partial x}(M_0) & \frac{\partial^2 f}{\partial z\partial y}(M_0) & \frac{\partial^2 f}{\partial z^2}(M_0)\end{pmatrix}$$

(4) Se det. val. proprii ale lui $H(M_0)$

$\det(H-\lambda I_3)=0$

Dacă toate val. proprii sunt strict pozitive, atunci M_0 este punct minim local

Dacă toate val. proprii sunt strict negative, atunci M_0 este punct maxim local

Dacă unele val. proprii sunt strict pozitive și altele strict negative, atunci M_0 nu e punct de extrem local

Dacă $\exists\ \lambda=0$, atunci se studiază semnul diferenței $f(x,y,z)-f(x_0,y_0,z_0)$, eventual cu ajutorul polinom. Taylor

Condițiile Sylvester

→ Dacă $\Delta_i>0,\ \forall i$ atunci M_0 este punct de minim local

→ Dacă $(-1)^i\Delta_i>0,\ \forall i$ atunci M_0 este punct de maxim local

→ Dacă $\exists\ \Delta_i=0$ atunci se studiază semnul diferenței $f(x,y,z)-f(x_0,y_0,z_0)$

→ Dacă nu e îndeplinită nicio condiție de mai sus atunci M_0 nu e punct de extrem local

⑥ Să se determine punctele de extrem local pentru funcția $f(x,y)=4x^2+\frac{2}{xy^2}+y^2$ $\qquad$ $x,y\neq 0$

$$\begin{cases}\frac{\partial f}{\partial x}=8x+\frac{2}{y^2}\left(-\frac{1}{x^2}\right)=0\\ \frac{\partial f}{\partial y}=\frac{2}{x}\cdot\left(-\frac{2}{y^3}\right)+2y=0\end{cases}\Rightarrow\begin{cases}4x=\frac{1}{x^2y^2}\\ y=\frac{2}{xy^3}\Rightarrow x=\frac{2}{y^4}\end{cases}\Rightarrow$$

$$\Rightarrow 4\cdot\frac{2}{y^4}=\frac{1}{\frac{4}{y^8}\cdot y^2}\Rightarrow\frac{8}{y^4}=\frac{y^6}{4}\Rightarrow y^{10}=32\Rightarrow y=\pm\sqrt{2}$$

$$x=\frac{2}{4}=\frac{1}{2}$$

$M_1(1/2;\sqrt{2})$

$M_2(1/2;-\sqrt{2})$

$$\frac{\partial^2 f}{\partial x^2}=8+\frac{2}{y^2}\cdot\frac{2}{x^3}$$

$$\frac{\partial^2 f}{\partial y^2}=\frac{4}{x}\cdot\frac{3}{y^4}+2$$

$$\frac{\partial^2 f}{\partial x\partial y}=\frac{4}{y^3}\cdot\frac{1}{x^2}$$

$$H(1/2;\sqrt{2})=\begin{pmatrix}24 & 4\sqrt{2}\\ 4\sqrt{2} & 8\end{pmatrix}\qquad \Delta_1,\ \Delta_2$$

$$H(1/2;-\sqrt{2})=\begin{pmatrix}24 & -2\sqrt{2}\\ -2\sqrt{2} & 8\end{pmatrix}$$

$M_1(1/2;\sqrt{2})$ minim local pt. că $\Delta_1,\Delta_2>0$

$\Delta_1=24$

$\Delta_2=24\cdot 8-16\cdot 2$

$M(1/2;\sqrt{2})$ minim local pt. că $\Delta_1,\Delta_2>0$